

Practice 2: PSD OF RANDOM SIGNALS

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https://github.com/santiagoUIS/GNURADIO_COMUIS_2025_2_B1_G7.git

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Abstract—The analysis of random signals is fundamental to understanding the behavior of communication, instrumentation, and data processing systems in noisy environments. In particular, power spectral density (PSD) allows us to describe how the energy of a signal is distributed in the frequency domain, providing a key tool for characterizing stochastic phenomena that cannot be analyzed solely in time due to the complexity this represents. This laboratory practice focuses on the analysis of the Power Spectral Density (PSD) using the GNURadio platform. In this work, different configurations of random binary signals, white noise, and real-world data sources such as images and audio were implemented. Additionally, the effects of interpolation, bandwidth, and variations of key parameters on the spectral shape of the signals were studied. The results obtained allow the comparison between theoretical principles and practical simulations, strengthening the understanding of the spectral behavior of digital signals and their applications in communication systems.

keywords: Energy, Noise, Random binary signals, Real-word data, Communication system, PSD.

I INTRODUCTION

The study of signals in the frequency domain was considered a fundamental aspect of communication theory, since it provided insights into how energy was distributed and transmitted through different media. The Power Spectral Density (PSD) served as a key tool that allowed a detailed analysis of this distribution, revealing hidden patterns and enabling the evaluation of transmission system performance[1]. In this laboratory, GNURadio was employed as a software-defined development environment based on block structures, which facilitated the design and analysis of flowgraphs for digital signal processing[2]. Throughout the practice, various random signals were generated and examined, both simulated and obtained from real-world sources such as images and audio. This approach made it possible to observe the

relationship between design parameters, time-domain behavior, and the resulting spectral representation. The main objective of this laboratory was to internalize the concepts related to the generation and analysis of random signals, while developing practical skills in the use of simulation and programming tools for digital signal processing. In doing so, the theoretical foundations of PSD and its application in modern communication systems were reinforced, where the understanding of spectral behavior played a crucial role in achieving efficient and reliable performance.

II METHODOLOGY

This section presents in an orderly and detailed manner the experiments carried out during the laboratory, including simulations and other considerations addressed within the practice.

II-A Analyze a bipolar random binary signal

The block diagram implemented in Fig. 1 uses different variables that parameterize the processing of the random input signal. Initially, a vector of $1M$ samples between $0[V]$ and $1[V]$ is generated. However, since we want to work with a signal with a mean of 0, $500[mV]$ is subtracted so that the signal has a mean of 0. The signal is then scaled by 2 to have discrete values of $-1[v]$ and $1[v]$. An interpolator filter is added to this, using a vector $\mathbf{h}$ of length $\mathbf{Sps}$ to simulate the increase in the **sampling rate** from its convolution with the bipolar signal. Finally, it is processed through a **Throttle** block to prevent the CPU of the device used from becoming saturated.

In addition, the block diagram in Fig.2 is implemented to obtain the PSD of the bipolar output signal and its respective visualization. Based on this simulation, the influence of the parameters that define the system are studied.

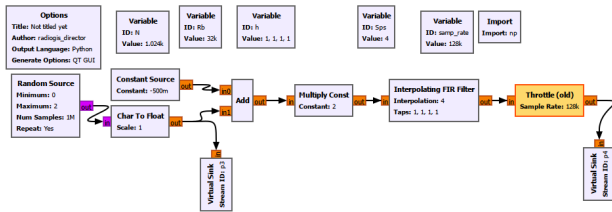


Figure 1. Block diagram of a bipolar random binary signal generator.

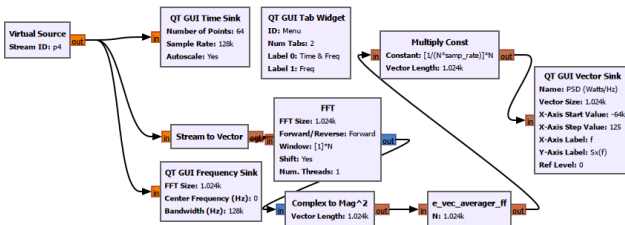


Figure 2. Block diagram that calculates and displays the spectral power density.

II-B Relationship between block diagram parameters

The analysis of a digital signal requires knowledge of the characteristics that determine its behavior over time and frequency. Parameters such as the bit rate (R_b), the sampling frequency ($samp_rate$), and the bandwidth (B) not only describe the signal quantitatively, but also have a direct relationship with each other.

The bit rate establishes the speed of information transmission and can be calculated based on

$$R_b = \frac{1}{T_b}, \quad (1)$$

where T_b is the bit time of the input signal. The sampling frequency defines the number of samples needed to represent the signal and can be obtained from

$$samp_rate = R_b \times Sps. \quad (2)$$

$$f_{s,p4} = S_{ps} f_{s,p3} \quad (3)$$

Finally, bandwidth determines the spectral space required for transmission without significant losses. In discrete communications, bandwidth is essential to avoid bit errors, whose frequency components are usually distributed across both low and high frequencies. Bandwidth is calculated as follows

$$B = \frac{2 \times samp_rate}{Sps}, \quad (4)$$

III EXPERIMENTS AND RESULTS

III-A Analysis of the block system based on a random bipolar input

To analyze the influence of system parameters, Sps is varied. Based on the mentioned equations, data such as bit rate 1, sampling frequency 2, and bandwidth 4 were collected. The results obtained can be seen in Table 1. Simulations of the output behavior for different Sps can be seen in Fig. 3.

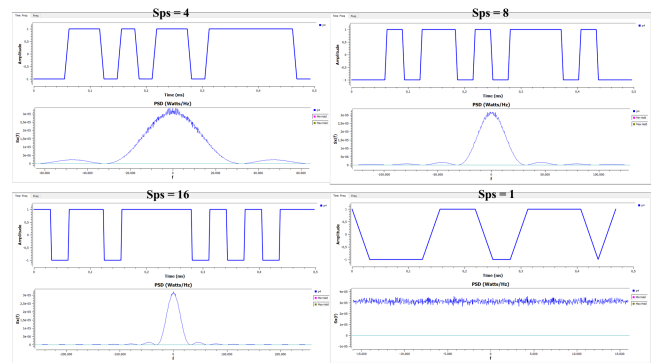


Figure 3. Time and frequency analysis of the bipolar output signal by varying the Sps parameter.

Table 1
RELATIONSHIP BETWEEN SPS, SAMPLING RATE, AND BANDWIDTH

SPS	R_b (kbit/s)	Sample rate (kHz)	B (kHz)
1	32	32	64
4	32	128	64
8	32	256	64
16	32	512	64

As the length varied, evidence was obtained that, for the function over time, when using the interpolator filter, the bandwidth required to obtain the signal is lower. This is because, in frequency, the total PSD corresponds to the product of the individual contributions of the SINC of each pulse. This is why it is said that the block simulates an increase in the sampling frequency, since the high-frequency components, which are not as recurrent as the low-frequency ones, gradually disappear when they interact with each other. Given that the bandwidths were kept constant for the experiment, due to the relationship described in eq. 4, it can be observed that for smaller values of Sps, the pulse shape is less well defined compared to a larger Sps.

Furthermore, for the case $Sps = 1$, it can be observed that the sampled signal actually behaves like a random noise signal. This is easy to corroborate by observing its PSD, which shows a flat floor throughout

the frequency domain. This can be explained from eq. 1, where $\lim_{T_b \rightarrow 0} R_b = \infty$. Due to the short bit time, the bandwidth B required to sample the bipolar signal must be ∞ , and since only a sampling frequency of 32k is available, the signal obtained does not provide the desired information.

III-B Analysis of the block diagram for noise input

Subsequently, the random bipolar input signal is replaced by noise to perform the respective time and frequency analysis. The simulations obtained for different values of Sps are shown in Fig. 4. It can be observed that the noise resolution increases as the size of the interpolator vector increases and, at the same time, the noise floor observed in the PSD graphs decreases.

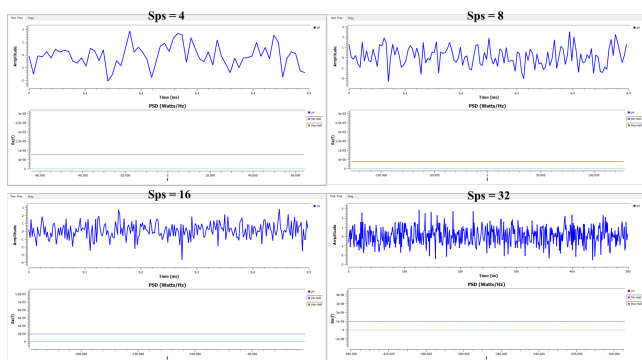


Figure 4. Time and frequency noise analysis based on variations in the length of the interpolator block vector.

By increasing the size of the vector used in the convolution operation, the block behaves like a filter with a broader response in the time domain, which translates into more selective filtering in the frequency domain. This greater selectivity attenuates high-frequency components where noise tends to concentrate much of its energy. Consequently, by increasing the length of the convolution vector, the total power of the noise present in the output signal decreases, which is reflected in a reduction in the noise floor level observed experimentally.

III-C Time and frequency response of the system for a real-world input

To avoid limiting the analysis to random signals generated by software, the noise input is replaced by real signals commonly used in communications systems, such as audio and images.

1) Image input experiment

In the block diagram, the image is first processed by a Unpack K Bits block with $K=8$, which takes each byte of the input signal and breaks it down into its 8 constituent bits, thus generating a pure binary sequence of zeros and ones. Next, the Char to Float block with a scale factor of 1 converts each character bit into a floating point value without modifying its amplitude, only adapting the data type to allow for further analysis, this visualization is achieved by changing the source displayed by P3 in Fig. 2. As a result of this processing, the temporary output shows a unipolar sequence of square pulses of amplitude 1 representing the bits of the image. In the frequency domain, the power spectral density reveals an almost flat spectrum typical of an unfiltered binary sequence, while the dB scale analysis shows a pronounced peak around zero frequency, corresponding to the continuous component produced by the imbalance between the number of ones and zeros in the data. All of this can be seen in Fig. 5.

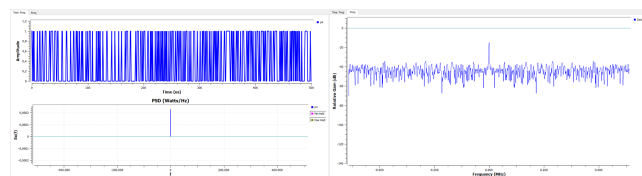


Figure 5. Display of input image with corresponding simulations of time, PSD, and frequency.

Once the input signal has been analyzed, it is processed using the block diagram in Fig. 1 performing the same Sps variations to study noise. The results obtained for the simulations can be seen in Fig. 6.

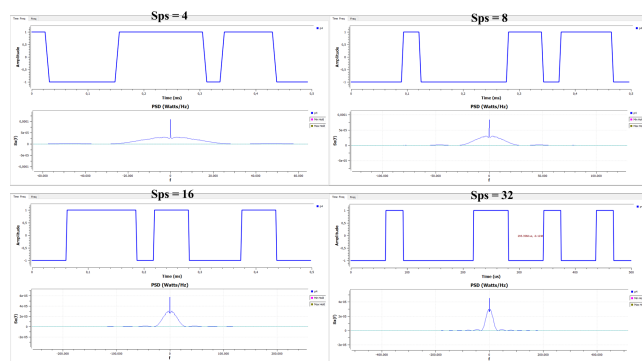


Figure 6. Two-dimensional signal processed with different Sps values.

Regarding the PSD analysis, it can be observed that, after filtering, the pulses are no longer ideal and change the power distribution of the signal, now appearing as

a sync and adding a DC component at 0 due to the bit imbalance.

2) Audio input experiment

The same process is repeated for the audio signal, initially studying its characteristics beforehand, and then after it is processed by the block diagram, observed in Figs. 78 respectively.

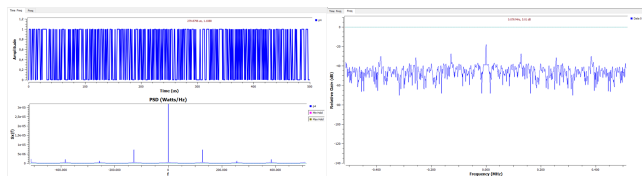


Figure 7. Display of input audio with corresponding simulations of time, PSD, and frequency.

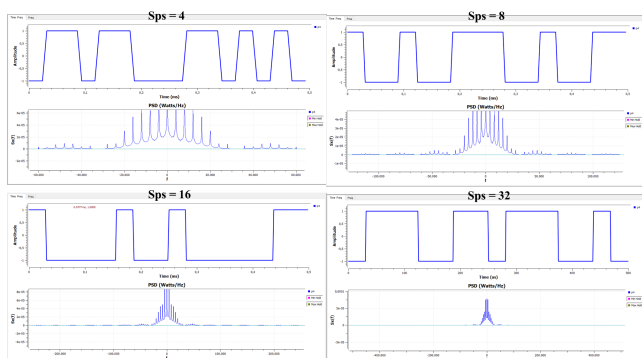


Figure 8. One-dimensional signal processed with different Sps values.

The PSD of the audio signal shows a typical distribution of sound content, with the highest concentration of energy located in the low frequencies and decreasing progressively towards the high frequencies. The spectrum shows well-defined peaks corresponding to characteristic audio tones or harmonics, while the rest of the band presents a relatively uniform noise floor representing random background content. This shape reveals a signal with a tonal structure and a limited effective bandwidth, in which the main energy is concentrated in a limited region of the spectrum and decreases continuously as the frequency increases.

IV CONTRIBUTIONS

The laboratory practice allowed establishing a direct relationship between the theoretical concepts of power spectral density (PSD) and their validation through simulations. The analysis of bipolar binary signals, noise, and real-world inputs such as audio and images showed how parameters like bit rate, samples per symbol (Sps), and bandwidth decisively affect the behavior of signals

both in the time and frequency domains. This process strengthened the understanding of fundamental concepts such as autocorrelation, pulse shaping, and spectral efficiency, which are essential in modern communication systems.

Moreover, the inclusion of real-world signals enriched the practice by moving the analysis beyond artificial signals into more practical scenarios. Audio signals exhibited a higher concentration of energy at low frequencies, while images displayed predominance at low spatial frequencies. These results reinforced the ability to interpret PSD in real contexts and demonstrated the importance of spectral analysis as a tool to characterize different types of signals.

V CONCLUSIONS

It was verified that the spectral shape of a random bipolar signal mainly depends on the transform of the pulse employed. In particular, with rectangular pulses the PSD presented the characteristic sinc^2 envelope, which was confirmed experimentally by observing how an increase in Sps improves the temporal definition of the pulse and reduces the high-frequency components. Before filtering, the bipolar sequence behaved as white noise with a flat PSD, while after shaping it adopted the expected theoretical structure.

When the input was replaced with noise, it was observed that a longer interpolation filter significantly reduced the noise floor in the PSD, acting as a temporal smoother and a more selective frequency filter. Similarly, the analysis of real-world signals confirmed these principles: audio concentrated its energy in low frequencies related to voice and instruments, while images concentrated in low spatial frequencies corresponding to large homogeneous regions, with fine details reflected in high-frequency components.

El método Welch demostró ser una herramienta adecuada para la estimación del PSD, aunque la precisión de los resultados depende en gran medida de la elección de la ventana, el tamaño de la FFT y el solapamiento. También se destacó la importancia de aplicar el remuestreo con antialiasing para evitar distorsiones, así como la conveniencia de realizar análisis fuera de condiciones en tiempo real para garantizar una mayor coherencia. En general, la práctica reforzó las habilidades en el procesamiento de señales digitales y consolidó la comprensión de los fenómenos espectrales aplicables a los sistemas de comunicación modernos.

Although both the binary signal from the image and that derived from the audio are represented by bipolar rectangular pulses, the statistical nature of the information that originates them produces different spectral

power densities. The image, with more redundant data regions and less frequent transitions, concentrates its energy in low-frequency components, while the audio, characterized by greater temporal variability and a more random bit distribution, generates a broader spectrum with a greater presence of high frequencies. This confirms that the pulse shape alone does not determine the PSD; it is the temporal distribution of the bits, defined by the information source, that conditions the final spectral content.

References

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